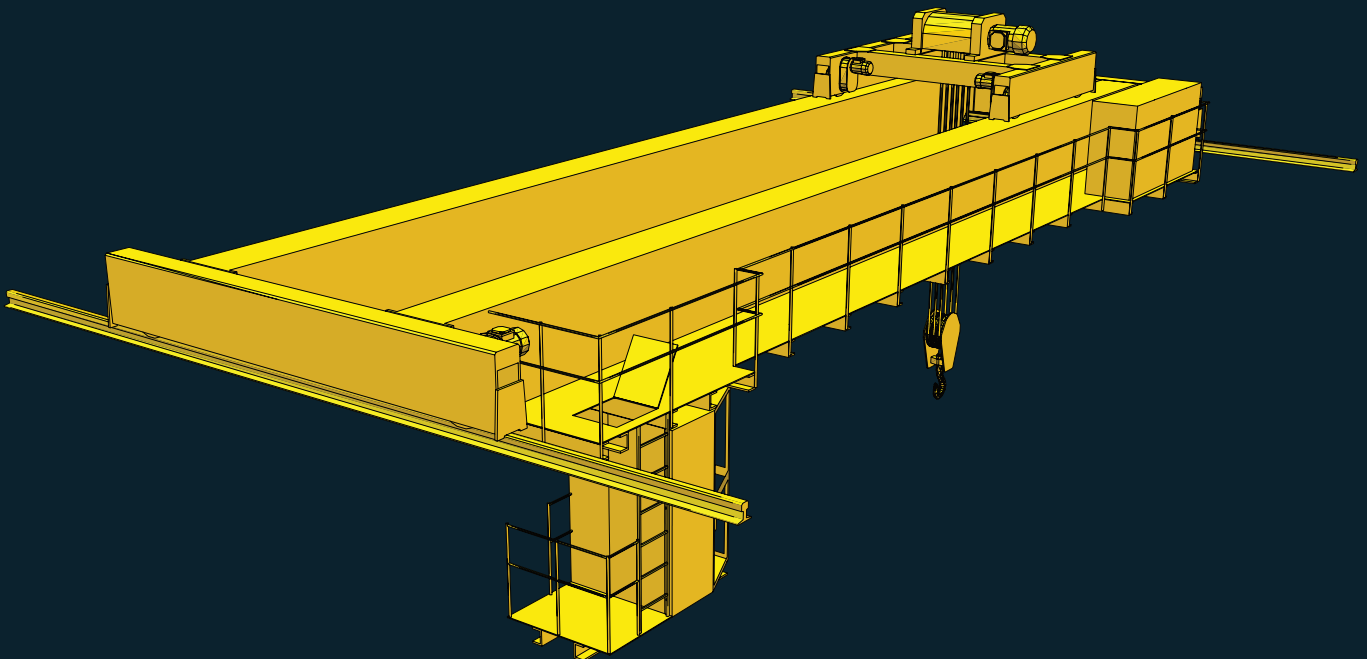


INNOMOTICS

INNOMOTICS MOVES! 1LE7

Crane Duty Motors

Optimized, Reliable and Sturdy



CRANE DUTY MOTORS:

Crane-duty motors are engineered for short, frequent starts/stops and intermittent duty cycles typical for cranes, hoisting, trolley and long-travel motions. They deliver high breakdown torque, withstand repeated thermal cycling and are suitable for VFD control for precise speed / positioning.

Performance Characteristics (S4 Duty Definition)

A sequence of identical duty cycles, each including a **significant period of starting**, a period of operation at constant load, and a rest and de-energized period.

Braking, if any, is mechanical and does not contribute to heating the windings.

t_r : The Cyclic Duration Factor (CDF) is the time during which motor heating takes place. It is given by:

$$t_r = \frac{t_A + t_B}{t_A + t_B + t_{St}}$$

Notes:

In S4 duty DOL operation, the effect of heating due to starting current is NOT IGNORABLE.

Salient Features:

Shaft height	71 to 180 FS in 4 & 6 pole (balance on request)
Power Range	0.37kW to 22kW in 4P 0.25kW to 18.5kW in 6P (balance on request)
Duty Type	S4 duty with 40% CDF, 150 SPH (different CDF,SPH on request)
Frame	Cast Iron
Efficiency Class	IE2 with S1 duty and standard output on main nameplate
Applicable Standards	Efficiency Classification according to IS 12615: 2018
Degree Of Protection	IP55 (IP56/IP65 options are available)
Voltages	415V (Standard) (Additional voltage variants available on request)
Frequency	50Hz
Type of construction	IMB3 (other construction variants available on request)
Cooling method	IC411 Totally enclosed fan cooled (IC416 available from FS 100 onwards)
Temperature class	155°C (F) utilized to 130°C (B) for line fed and 155°C (F) for VFD fed
Insulation system	VFD Suitable insulation scheme for voltage up to 480V as a standard.
Impulse Voltage Insulation Class (IVIC) for UN 415V	C (Severe) As per std. IEC 60034-18-41 / IS 15999 (Part 18/Sec 41): 2018.

Compliance & Nameplate Guidelines

- Efficiency classification as per IS 12615 for equivalent S1 outputs.
- Main nameplate: S1 duty data with S1 output and efficiency class.
- Auxiliary plate: S4 (intermittent duty) data — duty cycle, CDF, starts/hour, equivalent power, current and Mk/Mn
- S4 data can be moved to the main nameplate on request, however, S1 data and IE mark will be available on Auxiliary nameplate in accordance with IS 12615

Additional Information:

- Diagonally split terminal box as a standard
- Terminal box can be rotated through 360° in steps of 90°
- Terminal box mounting on RHS/LHS with L-shaped adaptor box enables easy interchangeability on site (applicable for frames 160 to 315)
- Increased cantilever force bearings at DE as an option

Ordering codes for Innomotics Moves! Crane Duty Motors 1LE7-IE2

Degree of Protection IP55, Insulation Class F, Ambient 50°C, Cast Iron housing, Method of Cooling - IC411, 415V ± 10%, 50Hz ± 5%, combined 10%, IMB3 (foot mounted) version as per IS:12615 / IEC:60034-1.

Line (DOL) operation (Temperature utilization limit to Class B)

4 - Pole 1500 rev/min							6 - Pole 1000 rev/min						
S1 duty kW		S4 duty kW 40% CDF, 150 SPH		Frame Size	Mounting Holes	Ordering code (MLFB)	S1 duty kW		S4 duty kW 40% CDF, 150 SPH		Frame Size	Mounting Holes	Ordering code (MLFB)
kW	HP	kW	HP				kW	HP	kW	HP			
240VΔ/415VY 50Hz+							240VΔ/415VY 50Hz+						
0.37	0.5	0.47	0.64	71	-	1LE7501-0CB32-3AA4-Z, Z=Y82	0.25	0.34	0.32	0.44	71	-	1LE7501-0CC32-3AA4-Z, Z=Y82
0.55	0.75	0.69	0.94	80	-	1LE7501-0DB22-3AA4-Z, Z=Y82	0.37	0.5	0.47	0.64	80	-	1LE7501-0DC22-3AA4-Z, Z=Y82
0.75	1	0.94	1.3	80	-	1LE7501-0DB32-3AA4-Z, Z=Y82	0.55	0.75	0.7	0.95	80	-	1LE7501-0DC32-3AA4-Z, Z=Y82
1.1	1.5	1.35	1.8	90S	S & L	1LE7501-0EB02-3AA4-Z, Z=Y82	0.75	1	0.9	1.2	90S	S & L	1LE7501-0EC02-3AA4-Z, Z=Y82
1.5	2	1.9	2.6	90L	S & L	1LE7501-0EB42-3AA4-Z, Z=Y82	1.1	1.5	1.5	2	90L	S & L	1LE7501-0EC42-3AA4-Z, Z=Y82
2.2	3	2.2	3	90L	S & L	1LE7501-0EB72-3AA4-Z, Z=Y82	415VΔ 50Hz						
415VΔ 50Hz							2.2	3	2.4	3.3	100L	-	1LE7501-1AC79-0AA4-Z, Z=Y82
3.7	5	3.7	5	100L	-	1LE7501-1AB79-0AA4-Z, Z=Y82+M1Y	3.7	5	3.9	5.3	112M	-	1LE7501-1BC79-0AA4-Z, Z=Y82
5.5	7.5	5.5	7.5	112M	-	1LE7501-1BB79-0AA4-Z, Z=Y82+M1Y	5.5	7.5	5.9	8	132M	S & M	1LE7501-1CC33-5AA4-Z, Z=Y82
7.5	10	7.5	10	132M	S & M	1LE7501-1CB23-5AA4-Z, Z=Y82	7.5	10	7.5	10	132M	S & M	1LE7501-1CC73-5AA4-Z, Z=Y82
9.3	12.5	9.3	12.5	132M	S & M	1LE7501-1CB79-0AA4-Z, Z=Y82+M1Y	7.5	10	8.2	11	160M	M & L	1LE7501-1DC23-5AA4-Z, Z=Y82
11	15	11	15	160M	M & L	1LE7501-1DB23-5AA4-Z, Z=Y82	11	15	11	15	160L	M & L	1LE7501-1DC43-5AA4-Z, Z=Y82
15	20	15	20	160L	M & L	1LE7501-1DB43-5AA4-Z, Z=Y82	15*	20	15	20	160L	M & L	1LE7501-1DC79-0AA4-Z, Z=Y82+M1Y
18.5	25	18.5	25	160L	M & L	1LE7501-1DB79-0AA4-Z, Z=Y82+M1Y	18.5*	25	18.5	25	180L	M & L	1LE7501-1EC79-0AA4-Z, Z=Y82+M1Y
22	30	22	30	180L	M & L	1LE7501-1EB43-5AA4-Z, Z=Y82							

VFD operation (Temperature utilization limit to class F)

4 - Pole 1500 rev/min							6 - Pole 1000 rev/min						
S1 duty kW		S4 duty kW 40% CDF, 150 SPH		Frame Size	Mounting Holes	Ordering code (MLFB)	S1 duty kW		S4 duty kW 40% CDF, 150 SPH		Frame Size	Mounting Holes	Ordering code (MLFB)
kW	HP	kW	HP				kW	HP	kW	HP			
240VΔ/415VY 50Hz+							240VΔ/415VY 50Hz+						
0.37	0.5	0.5	0.7	71	-	1LE7501-0CB32-3AA4-Z, Z=Y82	0.25	0.34	0.34	0.5	71	-	1LE7501-0CC32-3AA4-Z, Z=Y82
0.55	0.75	0.75	1	80	-	1LE7501-0DB22-3AA4-Z, Z=Y82	0.37	0.5	0.5	0.7	80	-	1LE7501-0DC22-3AA4-Z, Z=Y82
0.75	1	1.1	1.5	80	-	1LE7501-0DB32-3AA4-Z, Z=Y82	0.55	0.75	0.75	1	80	-	1LE7501-0DC32-3AA4-Z, Z=Y82
1.1	1.5	1.5	2	90S	S & L	1LE7501-0EB02-3AA4-Z, Z=Y82	0.75	1	1.1	1.5	90S	S & L	1LE7501-0EC02-3AA4-Z, Z=Y82
1.5	2	2.2	3	90L	S & L	1LE7501-0EB42-3AA4-Z, Z=Y82	1.1	1.5	1.5	2	90L	S & L	1LE7501-0EC42-3AA4-Z, Z=Y82
415VΔ 50Hz							1.5	2	2.2	3	100L	-	1LE7501-1AC42-3AA4-Z, Z=Y82
415VΔ 50Hz							415VΔ 50Hz						
2.2	3	2.9	3.9	100L	-	1LE7501-1AB43-5AA4-Z, Z=Y82	2.2	3	3.2	4.4	112M	-	1LE7501-1BC23-5AA4-Z, Z=Y82
3.7	5	4	5.4	100L	-	1LE7501-1AB73-5AA4-Z, Z=Y82	3.7	5	4	5.4	112M	-	1LE7501-1BC73-5AA4-Z, Z=Y82
5.5	7.5	6	8.2	112M	-	1LE7501-1BB73-5AA4-Z, Z=Y82	3.7	5	5.5	7.5	132S	S & M	1LE7501-1CC13-5AA4-Z, Z=Y82
5.5	7.5	7	9.5	132S	S & M	1LE7501-1CB03-5AA4-Z, Z=Y82	5.5	7.5	7.5	10	132M	S & M	1LE7501-1CC33-5AA4-Z, Z=Y82
7.5	10	9.3	12.5	132M	S & M	1LE7501-1CB23-5AA4-Z, Z=Y82	7.5	10	9.3	12.5	160M	M & L	1LE7501-1DC23-5AA4-Z, Z=Y82
11	15	14	19	160M	M & L	1LE7501-1DB23-5AA4-Z, Z=Y82	11	15	15	20	160L	M & L	1LE7501-1DC43-5AA4-Z, Z=Y82
15	20	18.5	25	160L	M & L	1LE7501-1DB43-5AA4-Z, Z=Y82	15	20	18.5	25	180L	M & L	1LE7501-1EC43-5AA4-Z, Z=Y82
18.5	25	22	30	180M	M & L	1LE7501-1EB23-5AA4-Z, Z=Y82							
22	30	27	37	180L	M & L	1LE7501-1EB43-5AA4-Z, Z=Y82							

S1 data will be punched on main nameplate with IE marking wherever applicable with S1 duty and standard output.

Respective S4 data will be punched on auxiliary nameplate. S4 data on main nameplate can be provided on request however S1 data and IE mark will be available on Auxiliary nameplate.

*Temperature utilization limited to class F (applicable for line operation)

-Z Y82 prices are inclusive in above LP.

M1Y is mandatory wherever applicable & prices are inclusive in above LP

+Ratings ≤1.5kW are star connected and ratings >1.5kW are delta connected (2.2kW/4P in Line operation is star connected)

INNOMOTICS MOVES! 1LE7

Voltage Code (Specified in MLFB Positions 12 & 13)

Frequency 50Hz						
Position 12 & 13	Connection		Position 12 & 13	Connection		Short Code
	Δ	Y		Δ	Y	
18	200V Δ		90	220V Δ	–	M1Y
20	–	360VY	90	230V Δ	–	M1Y
21	–	380VY	90	240V Δ	–	M1Y
22	–	400VY	90	360V Δ	–	M1Y
23	–	415VY	90	440V Δ	–	M1Y
27	–	500VY	90	460V Δ	–	M1Y
33	380V Δ	–	90	480V Δ	–	M1Y
34	400V Δ	–	90	525V Δ	–	M1Y
35	415V Δ	–	90	–	660VY	M1Y
40	500V Δ	–	90	–	690VY	M1Y
43	(575V Δ)	–				
46	660V Δ	–				
47	690V Δ	–				
90	Any other voltage					M1Y

Notes:

- Short codes are mandatory when 12 and 13 in MLFB is 9 and 0 respectively. • M1Y requires Hz, V and kW to be specified in plain text • For 60Hz please inquire

Innovation in motion

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